

CS 383 Fall 2026 – Asymptotic Analysis

Practice Questions (Ungraded)

1. Prove: $n \log n = \Omega(n)$

2. Compare growth rates of the following functions (i.e. if $f_1(n) \dots f_6(n)$ are the 6 functions stated below then You just need to arrange them in the order of growth. If $f_2(n)$ grows slowly than $f_1(n)$ but higher than $f_5(n)$ then the order would be $f_5(n) < f_2(n) < f_1(n)$. Arrange all 6 of them in ascending order)

$$n, \log n, n \log n, n^2, 2^n, n!$$

3. Prove or disprove:

Are the following True/False?

A. $n^2 = O(n^3)$

B. $n^3 = O(n^2)$

C. $5n + 100 = \Theta(n)$

D. $n \log n = \Theta(n)$

E. $2n^2 + 7n + 1 = \Theta(n^2)$

F. $\log(n^2) = \Theta(\log n)$